

MyBoeingFleet and Agile Software Development

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Abstract

This paper describes my experience in helping MyBoeingFleet.com moved to Agile over the last 24 months. The paper describes the existing environment, how CMMI became a potential hurdle, and how the Process group within Boeing became my surprise ally and an enabling factor of Agile at MyBoeingFleet and beyond.

1. Introduction

MyBoeingFleet.com is the leading extranet website used by The Boeing Company to provide customers of Boeing with access to technical documents, engineering drawings, parts, and numerous other information resources. It receives over 23 million hits a month.

2. A Brief History of IT Development

In 1996, when I joined Boeing, the official process for developing software was called P+, which leverages a software development lifecycle now called Macroscopic. The Macroscopic process tracks significant milestones by providing templates, typically in a Word document, for workers to fill in with details. The documents were part of a process where they would feed other documents further down the software development lifecycle. For example, a requirements document had to be completed prior to moving into the analysis series of documents. Because of the complexity of the process, a coach would typically accompany a team through the phases of development: Opportunity Evaluation, Preliminary Analysis, Architecture, Design and Construction, and Implementation. In 2005 this became Boeing's Application Development and Sustaining (AD&S) Process, which combined these phases with the Macroscopic documentation process. At the end of each phase, there is checkpoint, or gate, where stakeholders would approve the documentation set,

review the project process, and approve the team to move onto the next phase of the project.

This process met resistance, both active and passive, but also had adherents, who appreciated the rigor of the document structure and the formality of the process

3. Working Under the Radar

After a brief tour as an Electrical Engineer in 747 Landing Gear, I transferred into a group doing Java Development in 1998. In my first project as a software developer, I did not find value in the documents that I was supposed to use to guide my development. I was anxious to write code, and with Java moving into the forefront of languages I was flush with books and articles on the Internet. Many of my co-workers were senior developers, who did not feel that they required the guidance that the documents were providing, nor the associated process.

I cut my teeth in Java development for two years at that organization, learning mostly by experience and from expert mentors.

4. Get Rich Attempt

In 1999 the .com revolution was moving into high gear, and I attempted to participate in it by leaving Boeing and joining WRQ, a Seattle-based integrator made famous by the software that connects UNIX systems to PCs. This was the beginning of my education in Agile and iterative software development. Many of my co-workers were familiar with the Rational Unified Process and iterative development techniques. I was impressed at how rapidly this much smaller company moved and how important it was to them to change and adapt constantly. At WRQ I felt the competitive pressures from the marketplace make their way into my work statement and priorities. Very often developers were not held to any process but Design-Code-Test, which I will refer to as the No-Process option.

I was keenly aware of the Agile movement. It seemed to me that it provided the right amount of structure to the work. With the No-Process option, I observed that systems were difficult to maintain and extend. With the speed of development, systems were not refactored, and best practices were not followed. There were often either poor or non-existing architectures as systems grew organically. As the .com bubble was bursting, several senior folk left WRQ, either voluntarily or through layoffs. This meant doing more with less, and I inherited several systems created by the No-Process option. When I first joined WRQ, it was liberating to be free of heavy documentation and process. By the time of my exit from WRQ, it was clear that The No Process or what is now known as “Cowboy Coding” was not the right answer for delivering quality software, regardless of the talent and dedication of the people involved.

As I continued my career, I followed the development of Agile processes for seven years and eventually found that the closer the companies followed these processes and practices the more mature their code was. I saw the power of continuous integration, automated testing, iterative development, and a well-prioritized backlog.

Then, in 2006, I returned to The Boeing Company, this time as a Project Manager.

5. Back to the Past

Upon re-entering the Boeing workforce, I found that not much had changed with respect to their process. Waterfall still prevailed. Agile was not officially recognized or supported by Information Technology, and the word “Agile” seemed loaded with connotations of risk and Cowboy Coding. There were some efforts under different names, such as Rapid Prototyping or Accelerated Development. MyBoeingFleet.com was getting ready for a CMMI validation soon after my arrival in 2006.

Capability Maturity Model Integration (CMMI) is a process improvement approach that provides organizations with the essential elements of effective processes, and can provide a point of reference for appraising current processes.

MyBoeingFleet’s upcoming assessment meant that I was not able to do Agile. If I were to do a different process than the approved one, the assessor could potentially find issue with not following processes. I was informed that to pass this appraisal, MyBoeingFleet needed to institutionalize the existing Waterfall process by following it for a minimum of six months.

In hindsight, with more knowledge of CMMI and Macroscopic, it is clear to me that many of my peers and management within Boeing misunderstood these tools in fundamental ways that are difficult to correct. The rigor of the AD&S process was often misapplied in a rigid and dogmatic fashion, rather than in a more flexible way its creators designed.

6. Bringing Agility

I was disappointed by the lack of official Agile processes at Boeing. I knew that MyBoeingFleet.com, a smaller team that does web application programming, met the profile for Agile. I had to discover a way to quickly change the process prior to the upcoming CMMI validation. Fortunately there was a way to change the process. I was directed to submit a change request against the existing Waterfall process. I did so, but the change request languished in the queue of the process improvement team, who were less than excited about Agile. This process improvement team “owned” the process, and as such was dedicated to the existing process.

I found what I considered a loophole in Macroscopic 4.5 – a delivery path called “Accelerated,” which allowed iterative development after requirements were created. While not perfect, this gave me the cover to iterate, which was a step in the right direction.

I was able to score a few quick releases with my projects using this path. Unfortunately, the Accelerated project delivery path was mandated for smaller projects, less than 400 total hours. Since mine were much longer than, I was out of compliance. With the date of the CMMI evaluation approaching, the process improvement team pressed me to stop using unapproved processes. It was time for me to push for a management decision. I had to defend the change request and fight for Agile at MyBoeingFleet.

7. Showdown

In a series of rather intense meetings, those who appreciated the existing process presented their side, while I described by hoped-for change. With large printouts on the walls of a crowded conference room, most people sat silent as the debate raged on. Prior to this meeting, I had worked closely with first-level development managers for whom I had completed a few projects, and to their credit, they saw the value of Agile and had assisted me in its creation and during this debate. My approach was simple; I did not intend to tell anyone how to run projects. If there were Project Managers who were adamantly in favor of Waterfall, I

was not there to tell them to stop. I simply wanted permission to use Agile on my projects, and give others that option as well.

The process I was fighting for was essentially Scrum, which I thought had enough structure to succeed at Boeing. I also created a flowchart that showed the existing process with Agile, demonstrating an attempt at peaceful co-existence. This compromise was unfortunately rejected by the advocates of the Waterfall process. Despite the lack of concurrence, the change request was approved. Agile had obtained the go-ahead from first-line management and was sent to senior management for final approval.

8. Help From an Unexpected Ally

Obviously and thankfully, I am not the only Agile practitioner in Boeing IT. While I may have been more vocal than some, I was not alone. In a chance meeting, I was introduced to Ryan Kleps, who was one of the process people from our Software Quality and Compliance group. Ryan's group was responsible for creating the AD&S process, performing the CMMI audits (including the upcoming MyBoeingFleet audit), and other common processes improvements at Boeing. If Ryan or another assessor were to assess a group at a lower maturity level, say CMMI Level 2 when the group was attempting Level 3, it would mean a revamping of the development processes, and potentially a blot on everyone's end-of-year evaluation. Boeing takes process improvement **very** seriously.

It should be clear then why I was surprised to see Ryan as an Agile collaborator. Ryan and I attended Agile 2007 together and forged a friendship and a great working relationship that endures today.

The genesis of that working relationship was a workshop in November of 2006. The process group having created the AD&S, was aware of its shortcomings and sought to create an Agile alternative. Seeing my interest and experience, they involved me in its creation. Ultimately, the Agile process changes sent to senior management were approved, in part, due to the forward thinking and courageous work of Ryan and his organization. In December 2007 MyBoeingFleet was assessed at CMMI Level 3, including the Agile Process. After numerous conversations with Ryan and other CMMI personnel, it was apparent to me that there was much misunderstanding about the nature of CMMI. The core of CMMI, the Key Process Areas or KPAs, were achievable with a host of different methodologies. CMMI and the associated audits were not forcing a particular methodology, but attempting to help the

group follow best practices by asking penetrating questions. Is your process documented? Is it repeatable? Do people know about your process and follow it? Are you following software development and project management best practices?

9. Enterprise, Beam Me Up

With Agile AD&S now defined for all of Boeing IT, the Process group kicked off a pilot of about seven projects across Boeing, two of which were in my group. The pilot went for approximately four months and was completed with positive results almost across the board. Shortly afterwards, Agile AD&S was approved by executive management to roll out company-wide. I could now say the word "Agile" at Boeing.

It has been about six months since this approval. Ryan's team has created the Agile AD&S website using a wiki, which encourages participation, and emphasizes the openness and collaborative nature of the effort. There is a weekly Agile Support Group meeting that is well attended across the country. The goal of the support group is to enable the viral spread of Agile AD&S through collaboration. I have been active in Agile as well, giving presentations to groups both within my management structure and without. My personal strategy for continuing the momentum of Agile within Boeing is to focus on management. In a large company like Boeing, no single voice or individual approval can ensure an initiative's success. The Agile community in Boeing must show their respective layers of management the advantages in the appropriate use of Agile. The tough work is still ahead for the Boeing Agile community. There are several issues that we face:

- Few product owners and no access to customers.
- Small population of Agile practitioners.
- Matrixed organization – most knowledge workers I'm aware of are assigned to multiple projects, even across departments. Boeing has numerous centralized functions such as database administration, server administration, architecture boards, and project management.
- Few teams are fully co-located.
- Limited ability to customize workspace.
- Culture resistance is still fairly strong. Boeing in my view is generally risk-averse and values moving carefully above moving quickly. Boeing enjoys terrific employee retention rates, which means software practitioners are usually very experienced.

Bringing change to an expert community of seasoned employees is especially challenging.

- Lack of consistent sponsorship of Agile.
- Difficult to get good estimates.
- Difficult to obtain a useful velocity.

10. Lessons Learned

So, what lessons did I learn? What would I do differently?

- **Training.** I do not believe I would have been successful without having attended Scrum training prior to my adventure. While I had traditional project management training, and significant experience with Agile, it strengthened my resolve to have my Scrum Master Certification. Additionally, role-based training would have helped assist team members: Emergent Design classes for architects, test-driven development for developers, and product owner training for business analysts and product owners. This training would have aided the transition to Agile from a process that some did not want to leave by showing them the value proposition from a voice of experience.
- **Get A Tool.** Tools for Agile Project Management make it easier for new teams to see how the process is working.
- **Product Ownership.** Attempt to do projects with more active product owners in them. It is very difficult to perform Scrum-master/PM type of work and also be the produce owner. With a strong business perspective on your team, it makes it easier to resolve priorities on stories and craft a product strategy that the P.O. can own and evangelize.
- **Stick to the process.** I have started teams with what I call the three pillars of Agile: the Daily Standup, the Fixed-Duration Iteration with demonstrations of working code at the end, and a prioritized backlog as the primary vehicle for requirements and scope management. Do not discard Agile at the first sign of trouble. Chances are it is not the process that is causing it.
- **Maintain the current reporting structure, and then go beyond it.** Avoid irritating management by not doing things they ask you to do. If the need is to report Earned Value, do it, and then add the burndowns. That way you do not appear lazy or recalcitrant. Doing this also provides the opportunity to demonstrate

why a certain metric may communicate what the management needs better than the current methods.

- **Demonstrate.** Show working code. Demonstrations are usually fun, invite involvement from management, and show the rapid response of Agile.
- **Documentation.** Show your teams that Agile allows documentation. Give your teams a lean way to communicate, such as a wiki. Many of the current tools come with built in wikis that make this easy. By using an online tool like a wiki, I attempted to break the paradigm of the document-based project process.
- **Understand people.** Most people just want to do a good job. They have pride in their work and believe that they are good at what they do. Bring Agile to the team in a way that helps them understand why changes are needed, and highlight what they're doing right. Involve them in the change, but know that not everyone will be on board. Consensus is not always possible.

11. Conclusion

When you want to change people's minds and hearts, it's best to take the long view. The Boeing Company is over 150,000 people strong and has a long tradition of excellence. It is simply a matter of time before the value and power of Agile is understood and appreciated at the highest levels.

Having fun and letting your teammates and management see your passion will give them the confidence to follow you down this exciting, but often unnerving path towards greater Agility.

12. Acknowledgements

I'd like to thank Ryan Kleps and Herb Bowie from what I've been calling the Process Group, which is actually a fine organization called Software Quality and Compliance (SQ&C). Their dedication to Lean and Agile has made it a viable alternative within the enterprise.

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